Bidirectional-Aware Network Combining Transformer and Mamba for Hyperspectral Image Denoising

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***Abstract*— Hyperspectral images (HSIs) often suffer from various noises, such as Gaussian noise, stripe noise, impulse noise, and deadlines due to the influence of sensors and external environments. These noises significantly degrade the quality of HSIs and hinder subsequent applications. While most current Transformer-based methods can effectively remove certain types of noise, they struggle with wide stripe noise. In addition, Transformers are typically applied within local windows due to the limitation of computational complexity. Although window-shifting operations enhance the interaction between windows to a certain extent, this interaction remains insufficient for comprehensive global modeling. In light of these limitations, we propose a bidirectional-aware network combining Transformer and Mamba (BTMnet), which consists of bidirectional long–short distance attention (BLSDA) and channel-split Mamba (CSM). To better remove wide stripe noise, BLSDA is designed with two rectangular windows adapted to wide stripes in both vertical and horizontal directions, utilizing Transformers to compute attention relationships within windows and across different windows. To further integrate global information and enhance the interaction of features between adjacent windows, CSM extracts global features by scanning in four directions across different feature channels. In BLSDA, we applied bidirectional windows in vertical and horizontal directions, and in CSM, we conducted bidirectional scanning in vertical and horizontal directions. The combination of these techniques allows for the simultaneous extraction of bidirectional features from HSIs. By evaluating the metrics and visualization, the experimental results on simulated and real experiments prove that our method can achieve better results.**

***Index Terms*— Deep learning, hyperspectral image (HSI) denoising, Mamba, mixed noise, Transformer, wide stripe noise.**

I. Introduction

HYPERSPECTRAL images (HSIs) are applied in various fields because of their rich spectral information, such as classification [1], object detection [2], and tracking [3]. However, due to factors stemming from the instruments and environmental impacts during signal transmission, HSIs are inevitably affected by complex noise. The noises in HSIs obscure the true information of the images and hinder the image interpretation and subsequent applications [4].

The noises in HSIs can be classified into Gaussian noise, stripe noise, impulse noise, and deadlines. Among them, Gaussian noise arises from random variations in light intensity and follows a normal distribution. Stripe noise typically manifests as a series of bright or dark lines traversing the image horizontally or vertically. Impulse noise typically appears as isolated pixels with extremely high or low pixel values, caused by uneven responses from imaging sensors or errors in the transmission system. Deadlines may originate from damaged rows or columns in the image sensor, typically appearing as black lines where no information is captured. Unlike RGB and panchromatic images, HSIs experience noise in spatial and spectral dimensions, and the intensity and type of noise vary across different bands in HSIs. This complexity makes the task of HSIs denoising even more challenging.

In the past few decades, various methods have been proposed to address the issue of HSI denoising. These methods can be categorized into two groups: model-driven methods and data-driven methods. With clear physical interpretation, model-driven methods rely on mathematical models and provide a deeper understanding of the denoising process. Examples of such methods for HSI denoising include filtering, nonlocal mean (NLM) [5], [6], [7], [8], [9], total variation (TV) [10], [11], [12], [13], [14], sparse representation (SR) [15], [16], [17], and low-rank (LR) [18], [19], [20]. These methods are capable of capturing the fundamental principles of degradation processes and the characteristics of noise, demonstrating higher versatility across different sensors. However, due to the complexity of their models, these methods involve many parameters needing manual adjustment for various scenarios. Furthermore, these model-driven methods perform poorly when dealing with complex noise.

In recent years, data-driven methods have experienced rapid development and gained widespread popularity due to their exceptional performance and faster speed. These methods can be primarily categorized into convolutional neural network (CNN)-based methods [21], [22], [23], [24], [25], [26], [27], [28], recurrent neural network (RNN)-based methods [29], [30], and attention-based methods [30], [31], [32], [33], [34],

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[35]. Compared to model-driven methods, data-driven methods exhibit stronger nonlinear fitting capabilities, enabling them to learn complex nonlinear input–output relationships from vast hyperspectral datasets. This advantage allows them to harness their robust learning power to capture the essential characteristics of intricate noise, facilitating the construction of effective denoising models. CNN-based methods leverage learnable convolutional kernels to extract spatial and spectral correlations in hyperspectral data. However, the local receptive field of CNNs limits their capacity to extract long-range dependencies. Consequently, RNN-based methods emerged, which excel in modeling sequential data. However, the memory capability of RNNs for long sequences is constrained, resulting in suboptimal performance when dealing with extensive sequences.

Recently, Transformer-based methods incorporating self-attention have achieved remarkable results in both high-level and low-level tasks due to their ability to model long-range dependencies in HSIs. The requirement for HSI denoising places a high emphasis on detail recovery. Thus, Transformers employed for this purpose typically utilize smaller tokens to ensure the restoration of fine details, which implies that the image is unfolded into an extremely long sequence. However, the quadratic computational complexity of Transformers poses a challenge in terms of hardware computational limitations. To address this issue, researchers have designed Transformers with differently shaped windows and utilized window-shifting operations to enable global modeling. However, window-shifting operations have limited gains for global modeling and tend to cause unnatural transitions at window edges. Consequently, the challenge arises of how to simultaneously extract both local and global features.

Currently, scholars have also applied diffusion models [36] to HSI denoising and achieved promising results. Diffusion models offer a promising solution by leveraging their powerful generative capabilities and structured processing abilities. By training on high spectral image data, diffusion models can learn the underlying noise distribution and simulate the inverse diffusion process to progressively remove noise and restore clear images. Although diffusion models have strong generalization capabilities, their performance is not satisfactory when dealing with heavy mixed noise [37].

Mamba [38], which is based on the state-space model (SSM), has shone brightly in the field of computer vision [39], [40]. By utilizing linear attention, Mamba effectively extracts long-range dependencies within sequences, significantly addressing the issue of rapid memory growth during computer operations. This demonstrates Mamba's capability to model global features.

To achieve better interwindow interaction and local and global information extraction, this article proposes a bidirectional-aware network combining Transformer and Mamba (BTMnet), the network comprises bidirectional long–short distance attention (BLSDA) and channel-split Mamba (CSM). In BLSDA, we design vertical and horizontal rectangular windows tailored to the shape of wide stripes to remove wide stripe noise. The bidirectional window can extract features in horizontal and vertical directions of HSIs simultaneously. Furthermore, long–short distance attention (LSDA) is capable of simultaneously extracting attention relationships both within and across windows to achieve superior feature extraction. The Mamba approach in vision Mamba (Vim) [39] scans all channels of features in four directions, which is overly complicated for HSIs with numerous spectral and feature channels. To better fuse features from different directions, CSM employs a channel-split scanning approach, which extracts global information by scanning in different directions across different channel groups. The scanning algorithm also enhances the recovery of details at the edges of Transformer windows. We introduced vertical and horizontal windows in BLSDA and applied vertical and horizontal direction scanning in CSM. The combination of these two approaches realizes a bidirectional perception of the network. The main contributions of the proposed method are summarized as follows.

1) We propose BTMnet, which is capable of removing various types of noise, especially wide stripe noise from HSIs while achieving superior detail recovery.
2) BLSDA is introduced to extract local window features and the interactions between windows. The bidirectional window can extract features in horizontal and vertical directions of HSIs and remove wide stripe noise.
3) CSM is designed to model the global context. The channel-split scanning algorithm can extract features in vertical and horizontal directions and fuse features among different channels. Global scanning further enhances the edge details of the windows in BLSDA.
4) We conduct extensive experiments to validate the effectiveness of our proposed method. We compare its performance with several methods, using quantitative and visual evaluations. The results clearly demonstrate the superior performance of our approach.

The rest of this article is organized as follows. Section II provides a review of the related work on HSI denoising. In Section III, we present our network details. In Section IV, we demonstrate the effectiveness of BTMnet through both simulated and real-world experiments. Finally, we summarize the article and draw conclusions based on the experimental results in Section V.

II. Related Works

In this section, we review HSI denoising methods in two categories: model-driven and data-driven.

A. Model-Driven Methods

Model-driven optimization methods are a common and effective strategy in HSI denoising. These methods leverage prior information about the image and mathematical models to achieve denoising. Currently, the main types of model-driven methods utilized in HSI denoising include NLM, TV, SR, and LR.

In the early stages of HSI denoising, some methods [5], [6] applied 2-D image algorithms based on NLM sequentially to each band of HSIs. However, these denoising methods tailored

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for single-band images neglect the correlation between bands in HSIs, leading to significant spectral distortion. To address this issue, many researchers have proposed various methods that consider the spatial and spectral dimensions of HSIs. Maggioni et al. [7] introduced a 4-D extension of the BM3D filter, known as BM4D. Similarly, Qian and Ye [8] proposed the 3-D NLM, and Othman and Qian [9] further developed a 3-D wavelet method based on previous approaches. These methods take adjacent bands into account and achieve better denoising.

TV regularization methods are commonly employed to use the prior information of HSIs fully. TV regularization primarily smooths images by minimizing gradients while preserving edge information. For example, Yuan et al. [10] proposed the spectral–spatial adaptive hyperspectral TV (SSAHTV), considering both spectral and spatial differences. Takeyama et al. [12] proposed the hybrid spatial–spectral TV (HSSTV), which could better remove artifacts in HSIs. Chang et al. [11] proposed the anisotropic spectral–spatial TV (ASSTV) to smooth noise by modeling HSIs as a spatial–spectral volume. Peng et al. [14] proposed a fast model-based approach via a novel regularizer named the representative coefficient TV (RCTV) to simultaneously characterize the LR and local smooth properties of HSIs.

Some methods based on SR are also applied to HSI denoising. The SR-based method decomposes HSIs, preserving as many detailed features of the image as possible while removing noise. For example, Ye et al. [15] developed the multitask sparse nonnegative matrix factorization (MTSNMF) to integrate dictionary learning with sparse coding, and Lu et al. [16] developed a spectral–spatial adaptive SR (SSASR) to remove noise both in spatial and spectral dimensions.

The LR property of HSIs can be used for denoising. Zhang et al. [18] introduced the LR matrix recovery (LRMR) model tailored for mixed noise removal in HSIs. Furthermore, He et al. [19] presented the nonlocal meets global (NGmeet) model, which leverages the similarities across full-band patches. Zhuang et al. [20] utilized LR and self-similarity of HSIs to remove noise. Several approaches have also demonstrated promising results by integrating LR and TV. Notably, He et al. [41] proposed the TV-regularized LR (LRTV) matrix factorization model, incorporating TV regularization into the LR matrix decomposition framework. Additionally, Wang et al. [42] introduced the TV-regularized LR tensor decomposition (LRTDTV) to address both spatial and spectral smoothness.

Model-driven methods possess high interpretability, but their parameters need to be set and adjusted manually. The fitting ability of such methods in different scenarios is also limited, resulting in poor performance when removing mixed noise in various scenarios.

B. Data-Driven Methods

In recent years, data-driven methods have become increasingly popular in HSI denoising due to their ability to learn complex nonlinear mappings from large amounts of data automatically. The CNN has been widely used for HSI denoising due to its ability to extract neighborhood correlations. Yuan et al. [21] proposed the spatial–spectral deep residual CNN (HSID-CNN), which can extract the correlation between adjacent bands and achieve good results. Chang et al. [22] proposed a multichannel 2-D CNN named HSI-DeNet, which directly learns the 2-D filters in each layer. Bodrito et al. [23] presented a trainable spectral-spatial sparse coding (T3SC) model, which retains the fundamental sparse coding model while requiring a minimal amount of end-to-end data for training the network parameters. Xiong et al. [24] proposed the model-aided nonlocal CNN (MACNet) that combines spectral LR prior and spatial depth prior. Fu et al. [25] established the nonlocal spatial–spectral neural network (NSSNN), integrating spatial–spectral relationship, global spectral correlation, and nonlocal spatial correlation to extract features with more precise structures. With the increase in computing power, 3-D convolutions have been widely used in HSI denoising to extract spatial and spectral information simultaneously. The 3-D atrous denoising CNN (3DADCNN) proposed by Liu and Lee [27] employs 3-D dilated filters to expand the receptive field. Dong et al. [28] proposed a 3-D HSI denoising CNN (3D-DCNN), decomposing the 3-D filters as 2-D spatial filters and 1-D spectral filters. The RNN has also been used in HSI denoising, particularly for its strong ability to model spectral correlation in the spectral domain. For example, the 3-D quasi-RNN (QRNN3D) proposed by Wei et al. [29] combined the 3-D CNN with an RNN. These methods struggle to capture long-range dependencies due to the limitation of convolutional local receptive fields.

With the introduction of Vision Transformer (ViT) [43], an increasing number of studies have applied Transformers to HSI denoising. The local–global feature-aware Transformer-based residual (FATR) network proposed by Wang et al. [32] utilized a spectral embedding operation and a multiscale windows partitioning scheme to extract spectral–spatial features from the HSI. To extract features from both spatial and spectral dimensions, many studies have applied dual attention mechanisms to process the spatial and spectral dimensions of HSIs separately, such as the spatial–spectral Transformer (SST) [44] and the dual-stream Transformer (DSTrans) [34]. The 3-D quasirecurrent and Transformer (TRQ3D) proposed by Pang et al. [30] combines the advantages of CNNs and Transformers. A spectral-enhanced rectangle Transformer (SERT) proposed by Li et al. [35] simulates nonlocal spatial similarity using rectangular self-attention.

Current Transformers typically adopt smaller tokens to achieve the recovery of more details during HSI denoising, leading to an increase in sequence length after flattening the image into a sequence. Because the computational complexity of the Transformer grows quadratically with the sequence length, it requires extremely high computing power to model global images by the Transformer. Methods such as SERT [35] restrict self-attention within partial regions to avoid excessively long sequences, but inevitably result in the loss of global information. The emergence of Mamba has addressed the issue of computing power. Vim [39] applies Mamba to visual tasks and has achieved impressive results, significantly enhancing computational efficiency while maintaining compa-

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rable accuracy to Transformers. SSUMamba [45] proposed by Fu et al. conducts 3-D scanning of HSIs to extract the spatial and spectral information of HSIs and achieves the removal of complex noise. MambaIR [46] achieves impressive results in image denoising by introducing scanning in four directions. These works inspire us to utilize Mamba for HSI denoising tasks. By combining the window Transformer and CSM, our proposed BTMnet can achieve good denoising performance without requiring high computational power.

III. Methodology

In this section, we first introduce the preliminaries of Mamba, and then we provide an overview of the overall architecture of the proposed BTMnet. Subsequently, we elaborate on our proposed blocks consisting of BLSDA and CSM.

A. Preliminary: SSM and Mamba

The state-space model (SSM) is inspired by the continuous system, which maps a 1-D function or sequence $x(t) \in \mathbb{R}^L$ to $y(t) \in \mathbb{R}^L$ through a hidden state $h(t) \in \mathbb{R}^N$. A linear SSM can be represented as follows:

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$, $\mathbf{B} \in \mathbb{R}^{N\times 1}$, and $\mathbf{C} \in \mathbb{R}^{1\times N}$ are matrices that govern the relationships among the hidden state, the input sequence, and the output sequence. N represents the dimension of the hidden state.

Because computers can only process discrete data, (1) is typically discretized using zero-order hold (ZOH), which can be represented as follows:

$$\begin{aligned} \overline{\mathbf{A}} &= \exp(\Delta\mathbf{A}) \\ \overline{\mathbf{B}} &= (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A}) - I) \cdot \Delta\mathbf{B} \end{aligned} \tag{2}$$

where Δ represents the sampling time step, used to convert continuous parameters $\mathbf{A}$ and $\mathbf{B}$ into discrete parameters $\overline{\mathbf{A}}$ and $\overline{\mathbf{B}}$, respectively. After discretization, the discrete version (1) can be rewritten as follows:

$$\begin{aligned} h_t &= \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t + \mathbf{D}x_t. \end{aligned} \tag{3}$$

To accelerate the calculation, the recurrent form in (3) can be converted into the following convolution form:

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{M-1}\overline{\mathbf{B}}\right) \\ y &= x * \overline{\mathbf{K}} \end{aligned} \tag{4}$$

where M is the length of the input sequence x, $\overline{\mathbf{K}} \in \mathbb{R}^M$ is a structured convolutional kernel, and $*$ denotes the convolutional operation.

B. Overall Network Architecture

The HSI captured by remote sensing satellites is often contaminated by various types of noise, including Gaussian noise, impulse noise, stripe noise, and deadlines. The noise model can be formulated as follows:

$$Y = X + N \tag{5}$$

where $Y \in \mathbb{R}^{H\times W\times B}$ represents the noisy image, $X \in \mathbb{R}^{H\times W\times B}$ denotes the original image, and $N \in \mathbb{R}^{H\times W\times B}$ means complex noise. H, W, and B represent the height, width, and spectral channels of the HSI, respectively.

The overall structure of the proposed BTMnet is presented in Fig. 1. First, shallow features are extracted from the noisy HSI through a 3×3 convolution, yielding feature maps. After shallow feature extraction, the B spectral channels of the original image are expanded to C feature channels. Expanding the channel dimension is beneficial for enhancing the diversity and expressive capability of features. These feature maps then undergo three iterations of noise removal through Transformer and Mamba groups (TMGs), followed by a 3×3 deconvolution to obtain a clean HSI. We adopt 2-D convolutions at the beginning and end of the network. The first convolution is used to expand the channel dimension to enhance the diverse expression of features, while the second convolution serves to fuse multidimensional features. In the network, we employ a strategy of processing separately by channel, followed by merging and shuffling the channels. This network structure design incorporates the processing of HSI spectra to prevent spectral distortion in the denoising results. In feature processing, each TMG comprises three Transformer and Mamba blocks (TMBs), which are constructed by alternating connections of vertical TMBs (VTMBs) and horizontal TMBs (HTMBs). Serving as the fundamental unit in the network, TMB is structured with BLSDA, CSM, and multilayer perceptron (MLP). Next, we discuss each module in detail.

C. Bidirectional LSDA

The proposed BLSDA comprises vertical LSDA (VLSDA) and horizontal LSDA (HLSDA). Taking VLSDA as an example, the details of VLSDA are illustrated in Fig. 2. Initially, the input features F^t are equally divided into F_1^t and F_2^t along the channel dimension, where F_1^t undergoes short-distance attention (SDA) extraction and F_2^t undergoes long-distance attention (LDA) extraction. After computing the attention for F_1^t and F_2^t separately, the results are combined and then subjected to channel shuffling to obtain the final output. In channel shuffle operation, we first divide every four channels into a group, denoted as $C_O = 4 \times N$. Then, we rearrange the channels within each group, denoted as $C_S = N \times 4$. Compared to VLSDA, HLSDA differs only in the direction of window partitioning. A schematic of HLSDA window partitioning is shown in Fig. 3. The computation of VLSDA is carried out through

$$\begin{aligned} F_1^t, F_2^t &= \text{Split}\left(F^t\right) \\ \hat{F}_1^t &= \text{LDA}\left(F_1^t\right), \hat{F}_2^t = \text{SDA}\left(F_2^t\right) \\ \hat{F}^t &= \text{Shuffle}\left(\left[\hat{F}_1^t, \hat{F}_2^t\right]\right) \end{aligned} \tag{6}$$

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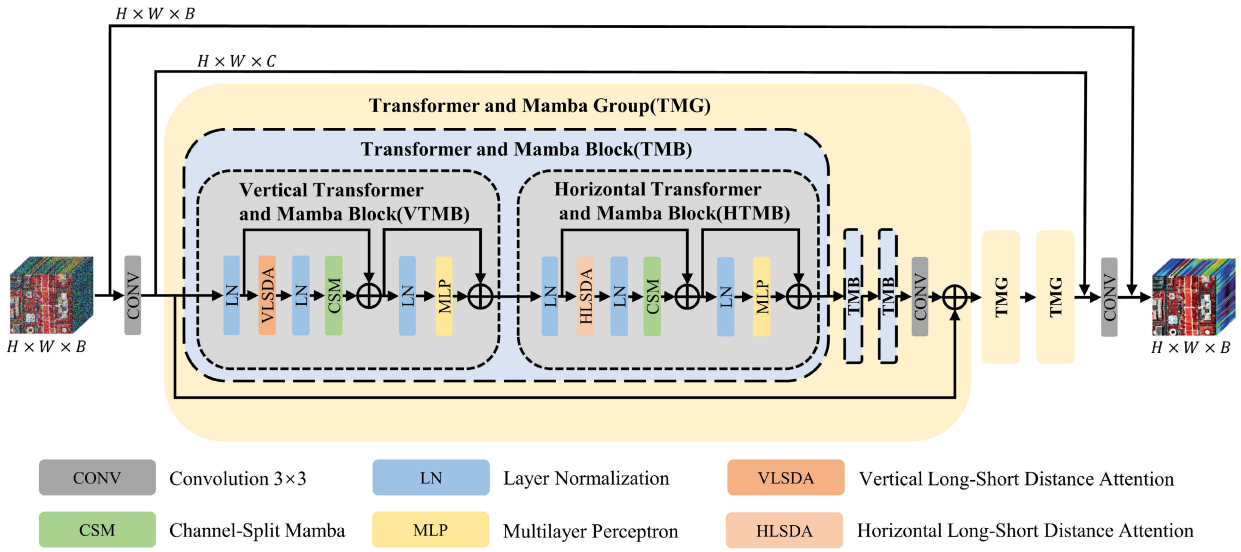


Fig. 1. Overall framework of the BTMnet.

where $F_1^t \in \mathbb{R}^{H\times W\times(C/2)}$ and $F_2^t \in \mathbb{R}^{H\times W\times(C/2)}$ are obtained by dividing the input feature $F^t \in \mathbb{R}^{H\times W\times C}$ along the channel dimension. SDA and LDA represent the mechanisms for capturing short-distance and long-distance dependencies, respectively. In SDA, F_1^t is further partitioned into N_1 windows of size $h \times w \times (C/2)$, denoted as $\{F_1^{t,1}, F_1^{t,2}, \ldots, F_1^{t,N}\}$, where h and w denote the height and width of the window, respectively, and $N_1 = (HW/hw)$. The attention operation for each feature window is represented as follows:

$$Q_1^i = F_1^{t,i} W_1^Q, \quad W_1^i = F_1^{t,i} W_1^W, \quad V_1^i = F_1^{t,i} W_1^V$$
$$\hat{F}_1^{t,i} = \text{Softmax}\left(\frac{Q_1^i {K_1^i}^T}{\sqrt{d}} + P\right) V_1^i \tag{7}$$

where $W_1^Q, W_1^K, W_1^V \in \mathbb{R}^{(C/2)\times(C/2)}$ are the linear projection matrix of $Q_1^i \in \mathbb{R}^{h\times w\times(C/2)}$, $K_1^i \in \mathbb{R}^{h\times w\times(C/2)}$, $V_1^i \in \mathbb{R}^{h\times w\times(C/2)}$. $P \in \mathbb{R}^{hw\times hw}$ is the positional encoding matrix, and d is the feature dimension.

For LDA, F_2^t is first partitioned into N_1 windows of size $h \times w \times (C/2)$, then these windows are reorganized into N_2 windows of size $(H/h) \times (W/w) \times (C/2)$, denoted as $\{F_2^{t,1}, F_2^{t,2}, \ldots, F_2^{t,N}\}$, where $N_1 = (HW/hw)$ and $N_2 = h \times w$. Following (7), these windows undergo attention operation and are then restored to their original shape.

D. Channel-Split Mamba

We introduced the CSM, as shown in Fig. 4, to extract global information and restore the details at the window edges in BLSDA.

First, the input features F^m are equally divided into F_1^m, F_2^m, F_3^m, and F_4^m along the channel dimension. Subsequently, each part undergoes scanning in a specific direction through Mamba block (MB): from top-left to bottom-right horizontally, from bottom-right to top-left horizontally, from left to right to bottom-right vertically, and from bottom-right to top-left vertically. In the different scanning directions shown in Fig. 4, Ⓢ represents the starting point of the scan and Ⓔ represents the ending point of the scan. Scanning in both horizontal and vertical directions enables CSM to perceive bidirectional information. Finally, the scanned results are concatenated along the channel dimension, and the channels are shuffled to obtain the final output $\hat{F}^m$. The computation of VLSDA is carried out through

$$F_1^m, F_2^m, F_3^m, F_4^m = \text{Split}(F^m)$$
$$\hat{F}_i^m = \text{MB}(F_i^m)\{i = 1, 2, 3, 4\}$$
$$\hat{F}^m = \text{Shuffle}([\hat{F}_1^m, \hat{F}_2^m, \hat{F}_3^m, \hat{F}_4^m]) \tag{8}$$

where $F_1^m \in \mathbb{R}^{H\times W\times(C/4)}$, $F_2^m \in \mathbb{R}^{H\times W\times(C/4)}$, $F_3^m \in \mathbb{R}^{H\times W\times(C/4)}$, and $F_4^m \in \mathbb{R}^{H\times W\times(C/4)}$ are obtained by dividing the input feature $F^m \in \mathbb{R}^{H\times W\times C}$ along the channel dimension. The structure of MB is illustrated in Fig. 4 and can be represented as follows:

$$y_1 = \text{SSM}(\text{SiLu}(\text{Conv1D}(\text{Linear}(\text{Norm}(x)))))$$
$$y_2 = \text{SiLu}(\text{Linear}(\text{Norm}(x)))$$
$$y = \text{Linear}(y_1 \otimes y_2) \tag{9}$$

where SSM stands for the state-space model, Conv1D represents a specially designed 1-D convolutional layer, $\otimes$ denotes multiplication, x is the input of the SSM, and y is the output of the SSM.

IV. Experimental Results and Analysis

In this section, we demonstrate the superior performance of the proposed BTMnet through comparison with other HSI denoising methods in simulated and real experiments. We also prove the effectiveness of each module and determine the optimal hyperparameters through ablation experiments.

A. Experimental Setting

1) Datasets: In simulation experiments, we use the Washington DC (WDC) Mall dataset. It is collected by the hyperspectral digital imagery collection experiment (HYDICE)

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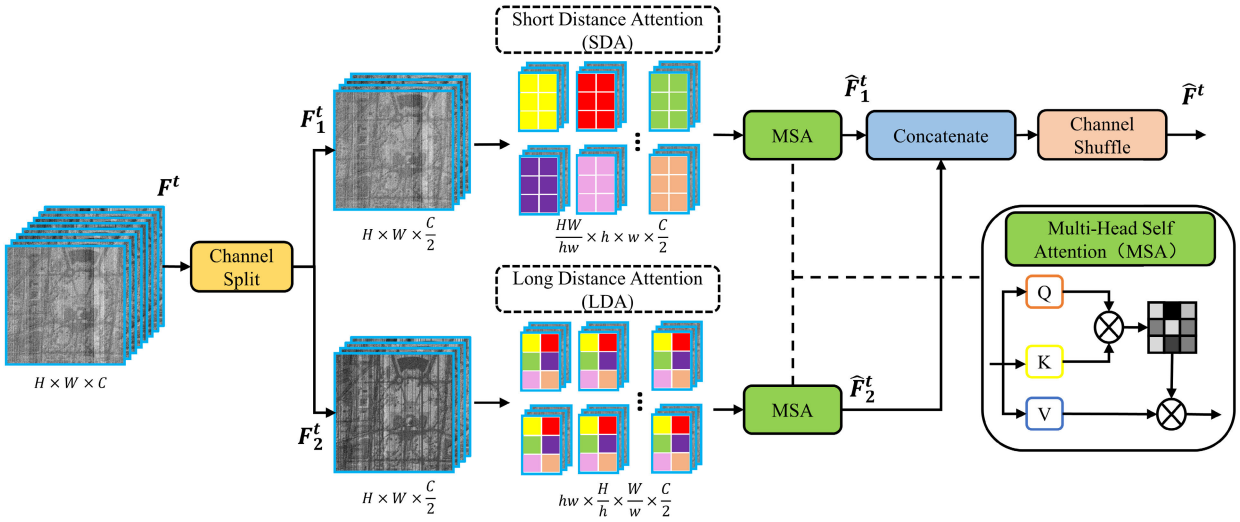


Fig. 2. Overall framework of VLSDA in BLSDA.

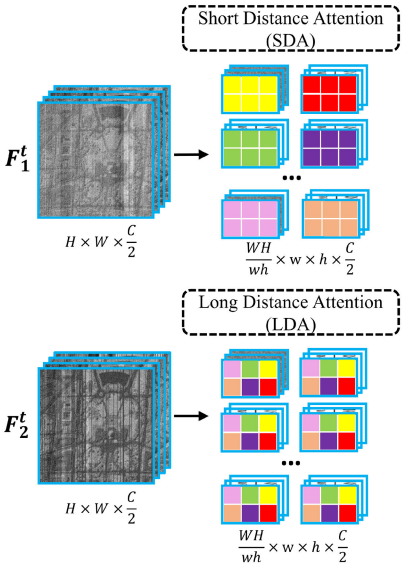


Fig. 3. Window partitioning framework of HLSDA in BLSDA.

sensor. The size of the image is 1208 × 307 pixels and it consists of 210 spectral bands. However, only 191 bands are considered for analysis after eliminating noisy bands. After duplicating one channel, the size becomes 1208 × 307 × 192. We then crop out an image with dimensions of 951 × 307 × 192 as the training set. When slicing the training set, to avoid redundancy in spectral information, we adopted a spectral grouping strategy. The spatial overlap rate is 50%, and the spectral dimension has no overlap. The size of the training image patches used in our experiments is 64 × 64 × 32, and a total of 8868 patches are used for training. For testing, we divided data of size 256 × 256 × 191 into six patches of size 256 × 256 × 32.

In real experiments, we selected four satellite datasets that are affected by mixed noise, especially stripe noise.

1) Zhuhai-1 (ZH-1) dataset, which is affected by wide strips and Gaussian noise and cropped to 256 × 256 × 32 for testing.
2) Gaofen-5 (GF-5) dataset, which is affected by periodic wide strips and Gaussian noise and cropped to 256 × 256 × 160 for testing.
3) Earth observing-1 (EO-1) Hyperion dataset, which is affected by complex noise and cropped to 384 × 192 × 166 for testing.
4) HYDICE Urban dataset, which is affected by complex noise and cropped to 256 × 256 × 210 for testing.

2) Quantitative Evaluation Metrics: To assess the effectiveness of the proposed method, we employ three frequently utilized evaluation metrics: the mean peak signal-to-noise ratio (mPSNR) [47], mean structural similarity index (mSSIM) [48], and spectral angle mapper (SAM) [49]. The mPSNR evaluates the quality of the reconstructed image by calculating the mean square error (mse) between the original and reconstructed images. A higher mPSNR score signifies superior image quality. mSSIM quantifies the structural similarity between the original and reconstructed images in terms of luminance, contrast, and structure. A higher mSSIM score indicates a closer structural resemblance. The SAM assesses the spectral similarity by calculating the angle between the pixel vectors of the two images. Lower SAM scores indicate a higher degree of spectral similarity. The definitions of these metrics are as follows:

$$\mathrm{mPSNR} = \frac{1}{C}\sum_{c=1}^{C} 10 \cdot \log_{10}\left(\frac{R^2 \cdot W \cdot H}{\sum_{i=1}^{W}\sum_{j=1}^{H}\left(X_c(i,j) - \hat{X}_c(i,j)\right)^2}\right) \tag{10}$$

$$\mathrm{mSSIM} = \frac{1}{C}\sum_{c=1}^{C} \frac{\left(2\mu_{X_c}\mu_{\hat{X}_c} + S_1\right)\left(2\sigma_{X_c\hat{X}_c} + S_2\right)}{\left(\mu_{X_c}^2 + \mu_{\hat{X}_c}^2 + S_1\right)\left(\sigma_{X_c}^2 + \sigma_{\hat{X}_c}^2 + S_2\right)} \tag{11}$$

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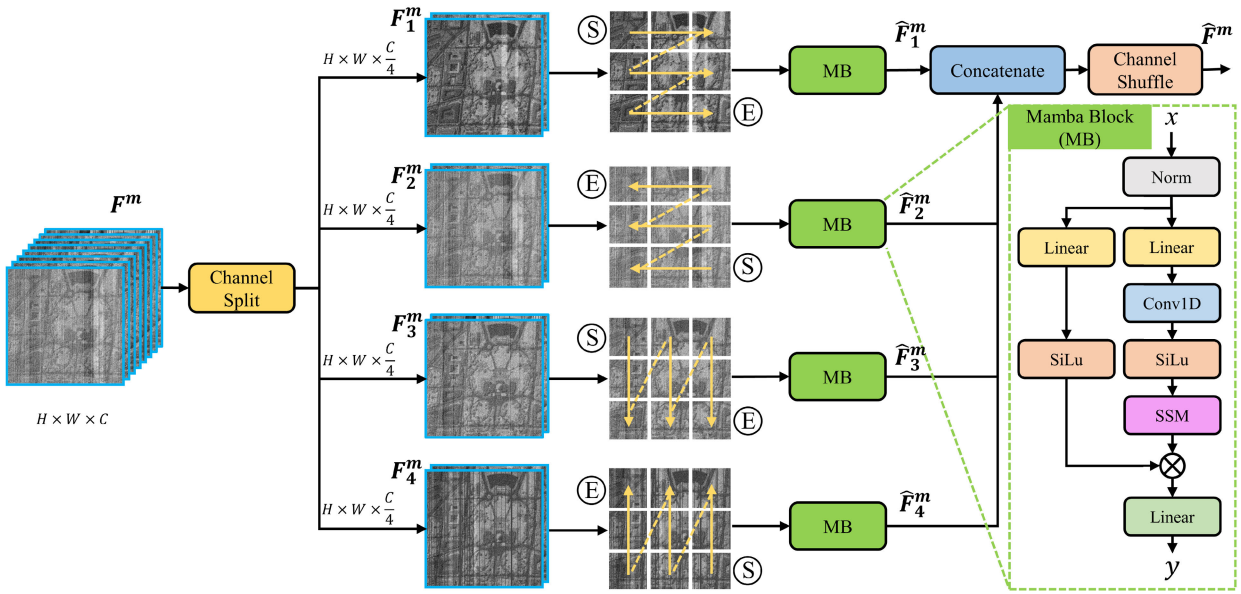


Fig. 4. Overall framework of the CSM.

$$\text{SAM} = \arccos\left(\frac{a \cdot b}{\|a\|\|b\|}\right) \tag{12}$$

where R represents the maximum possible pixel value. $X_c(i, j)$ and $\hat{X}_c(i, j)$ indicate the pixel values of the clean and processed images in channel c. μ_{X_c} and $\mu_{\hat{X}_c}$ denote the mean values of the clean and processed images in channel c. $\sigma_{X_c\hat{X}_c}$ represents the covariance between the clean and processed images in channel c. σ_{X_c} and $\sigma_{\hat{X}_c}$ denote the variances of the clean and processed images in channel c. S_1 and S_2 are constants introduced to stabilize the division. a represents the spectral vector of the clean image and b denotes the spectral vector of the processed image. Additionally, $||a||$ and $||b||$ denote the magnitudes of the spectral vectors for the clean and processed images.

3) Comparison Methods: We compare the proposed BTMnet with ten denoising algorithms, including LRMR [18], LRTV [41], LRTDTV [42], GRNET [50], MACNET [24], QRNN3D [29], T3SC [23], DSTrans [34], SST [44], and SERT [35]. The first three are model-driven methods and the last seven are data-driven methods.

4) Simulation Experiments: For the training dataset, we introduced various types of noise into each spectral band. Specifically, Gaussian noise with a standard deviation ranging from 0 to 75 is added to each band. Additionally, 33% of the bands are subject to impulse noise, with an intensity varying between 0.01 and 0.1. Another 33% of the bands contain deadlines, with density ranging from 5% to 15%. Similarly, 33% of the bands exhibit thin stripe noise, which has an intensity between 0 and 0.5 and occupies 5%–50% of the image width. Lastly, 33% of the bands are affected by wide stripes, with widths ranging from 5 to 25 pixels and intensity between 0 and 0.5, collectively occupying 5%–75% of the image width. The diversity in noise types and intensities ensures that the training dataset covers a wide range of challenging scenarios commonly encountered in practical applications.

The randomness of noise in real HSIs manifests itself in several aspects: random intensity, random density, and random frequency bands. To make the experimental results more authentic and credible, we randomly added noise to the WDC test dataset through the following settings.

1) *Case 1 (Gaussian Noise):* Gaussian noise is added to each band.
2) *Case 2 (Gaussian Noise + Thin Stripe Noise):* Based on *Case 1*, we add thin stripe noise to 33% of the bands.
3) *Case 3 (Gaussian Noise + Thin Stripe Noise + Deadlines):* Based on *Case 2*, we add deadlines to 33% of the bands.
4) *Case 4 (Gaussian Noise + Thin Stripe Noise + Deadlines + Impulse Noise):* Based on *Case 3*, we add impulse noise to 33% of the bands.
5) *Case 5 (Gaussian Noise + Thin Stripe Noise + Deadlines + Impulse Noise + Wide Stripe Noise):* Considering the presence of wide striping noise in remote sensing HSIs, and based on *Case 4*, we add wide stripe noise to 33% of the bands.
6) *Case 6 (Gaussian Noise + Bidirectional Thin Stripe Noise + Bidirectional Deadlines + Impulse Noise + Bidirectional Wide Stripe Noise):* Based on *Case 5*, we add bidirectional stripes and deadlines.

B. Training Details

1) Loss Function: To optimize our network, we use the mean square error (mse) to quantify the difference between the denoised HSI and the ground truth, which can be denoted as follows:

$$\mathcal{L} = \frac{1}{N}\sum_{i=1}^{N}\|\text{BTMnet}(\boldsymbol{Y}_i) - \boldsymbol{X}_i\|_F^2 \tag{13}$$

where $\boldsymbol{Y}_i$ and $\boldsymbol{X}_i$ denote the pair of noisy and clean HSIs, respectively. N represents the number of training patches in a batch.

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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON WDC MALL. THE BEST INDEX IS PLOTTED IN RED, AND THE SECOND-BEST INDEX IS PLOTTED IN BLUE

	Indexes	NOISE	LRMR	LRTV	LRTDTV	GRNET	MACNET	QRNN3D	T3SC	DSTrans	SST	SERT	BTMnet
Case1	mPSNR↑	19.45	28.79	28.79	24.55	28.37	33.79	27.86	30.30	29.50	33.10	33.61	35.60
	mSSIM↑	0.5832	0.9134	0.9182	0.8365	0.9389	0.9744	0.9231	0.9459	0.9435	0.9698	0.9735	0.9824
	SAM↓	28.87	8.71	9.74	6.54	7.10	5.12	9.15	6.99	6.87	5.59	4.85	4.03
Case2	mPSNR↑	17.27	26.83	28.16	24.12	27.52	31.89	26.77	29.58	28.75	31.69	32.64	34.53
	mSSIM↑	0.5263	0.8820	0.9059	0.8217	0.9288	0.9637	0.9073	0.9404	0.9375	0.9621	0.9684	0.9785
	SAM↓	31.83	10.86	10.13	6.89	7.60	5.83	9.78	7.48	7.20	6.12	5.07	4.30
Case3	mPSNR↑	16.21	24.85	27.50	23.73	27.58	29.89	26.30	28.56	28.54	31.84	32.54	34.59
	mSSIM↑	0.4973	0.8600	0.9053	0.8136	0.9222	0.9560	0.9049	0.9300	0.9331	0.9592	0.9667	0.9771
	SAM↓	33.77	12.36	11.10	6.93	7.92	7.17	9.85	8.42	7.37	6.64	5.31	4.43
Case4	mPSNR↑	15.61	24.75	27.53	23.72	27.50	29.51	25.78	28.06	28.25	31.98	32.94	34.86
	mSSIM↑	0.4711	0.8542	0.8978	0.8112	0.9151	0.9518	0.8974	0.9211	0.9269	0.9630	0.9700	0.9793
	SAM↓	34.47	11.96	10.45	6.95	7.99	6.89	9.81	8.74	7.50	6.48	4.98	4.19
Case5	mPSNR↑	14.79	22.07	24.57	22.86	27.10	28.18	25.74	26.85	27.94	31.60	31.73	34.21
	mSSIM↑	0.4559	0.8074	0.8914	0.7909	0.9112	0.9401	0.8992	0.9066	0.9237	0.9601	0.9626	0.9747
	SAM↓	36.93	18.24	10.18	8.72	8.57	8.22	9.94	10.67	7.93	7.03	5.71	4.68
Case6	mPSNR↑	11.74	18.78	23.42	19.82	25.19	24.13	23.91	24.38	26.02	28.29	29.90	30.95
	mSSIM↑	0.3584	0.7352	0.8787	0.6975	0.8813	0.8774	0.8617	0.8736	0.8977	0.9300	0.9477	0.9583
	SAM↓	41.75	23.62	11.27	11.15	10.57	11.73	11.33	13.66	9.62	8.31	6.81	6.19

2) Implementation Details: Among the three TMGs, we designed the window sizes as (64, 4), (64, 8), and (64, 16). This particular set of window shapes has demonstrated superior performance for our network. As for optimization, our initial learning rate is 0.0001, which decays to 50%, 25%, and 10% of the original learning rate at epochs of 10, 40, and 80, respectively, for a total of 100 epochs. Our training process is conducted on a Linux system with an NVIDIA A5000 GPU with 24 GB of memory. The specific environment configuration includes Python version 3.8.5, PyTorch version 1.11.0 + cu113, and CUDA version 11.3. All model-driven methods are implemented on a Windows 11 system using MATLAB R2023, which runs on an Intel[1] Core[2] i7-12700F CPU.

C. Simulation Experiments

1) Quantitative Results: We conducted simulation experiments on the WDC dataset for mixed noise removal, and the quantitative results are presented in Table I, with the best results highlighted in red and the second-best results displayed in blue. The noise types and intensities differ across the six cases due to the varying distributions of random noise. We have sorted the cases from 1 to 6 based on increasing noise levels. Cases 1–5 are designed based on different types and intensities of noise present in real HSIs, with Case 5 encompassing all types of noise and featuring the highest noise level. Case 6 is a simulated test set where we have added bidirectional noise to Case 5, which is used to show the performance of BTMnet in the presence of bidirectional complex noise. Overall, our proposed method outperforms the comparison methods in nearly all scenarios, achieving the best results.

In Case 5, BTMnet achieves the best results among all comparison methods in terms of the three metrics. The mPSNR and mSSIM values are significantly higher than those of other methods, demonstrating that our proposed network can better handle discrete random noise and structural noise. The lower SAM compared to other methods proved the effectiveness of the channel-split processing strategy. Additionally, it is observed that the data-driven methods outperformed the model-driven methods, highlighting the powerful nonlinear fitting capabilities of neural networks in addressing complex problems. In Case 6, our BTMnet continued to perform optimally, indicating that its ability to handle bidirectional noise is also superior to other methods. This further validates the superiority of the bidirectional window structure design in our network.

2) Qualitative Results: We demonstrate the qualitative results using the strongest noise case (Case 5) in Fig. 5. The noisy image contains Gaussian noise, impulse noise, thin stripes, deadlines, and wide stripes. Among the model-driven methods, LRMR can only remove part of the Gaussian noise, LRTV suffers from severe spectral distortion, and LRTDTV restores an overly smooth image, resulting in the loss of most details. Among the data-driven methods, GRNET and MACNET exhibit noticeable residual stripes. To compare the detailed textures of the restored images, we provide a zoomed-in local figure. In the enlarged red box, it can be observed that our proposed BTMnet excels in restoring the textures on the top of the building. In contrast, the line textures on the top of the building in the results of QRNN3D, T3SC,

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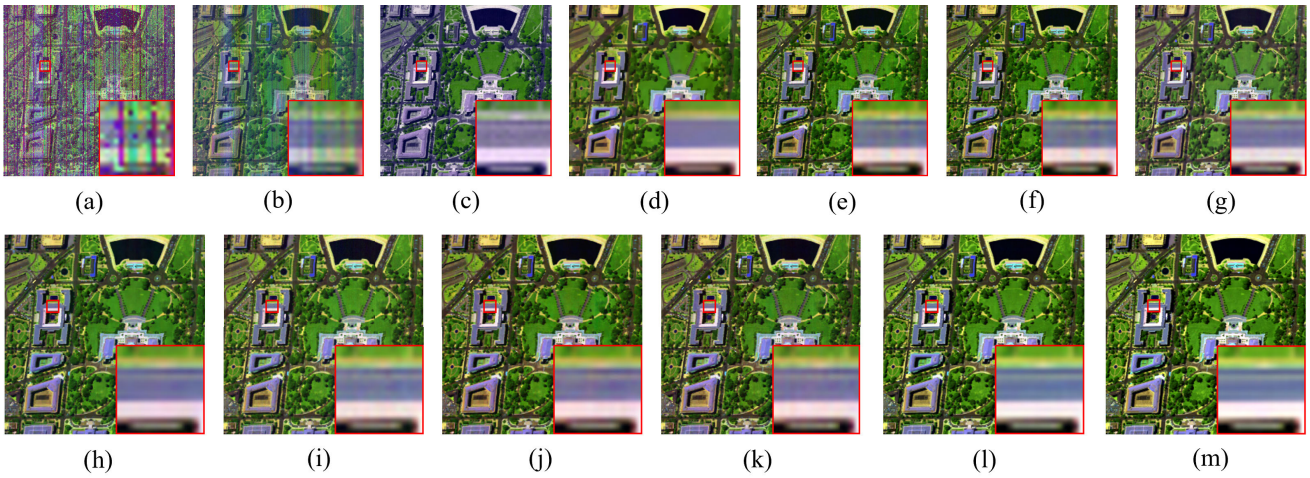


Fig. 5. Denoising results for the WDC dataset (Case 5) in the simulated data experiment with bands (171, 114, 15). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet. (m) Ground truth.

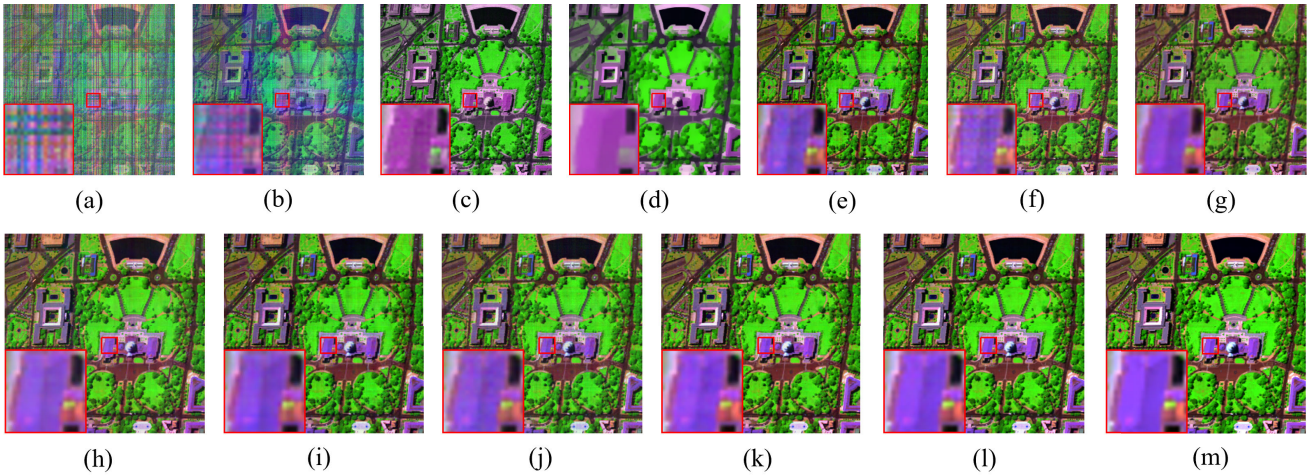


Fig. 6. Denoising results for the WDC dataset (Case 6) in the simulated data experiment with bands (152, 70, 24). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet. (m) Ground truth.

DSTrans, SST, and SERT appear blurred, potentially due to incomplete removal of stripe noise.

Fig. 6 presents the visualization results for Case 6. In Case 6, the noise intensity and type are consistent with those in Case 5, but the stripe noise and deadline noise are simultaneously present in both horizontal and vertical directions. This bidirectional noise exhibits a stronger destructive effect on the HSI structure. LRMR can only remove part of the Gaussian noise. LRTV, LRTDTV, and QRNN3D suffer from spectral distortion. GRNET and MACNET fail to completely remove the bidirectional stripes, exhibiting some residual stripe artifacts. T3SC, DSTrans, SST, and SERT fall behind BTMnet in terms of detail restoration. To compare the detailed textures of the denoised images, we provide enlarged partial views of the building area. In the enlarged images, our method most closely approximates the label image in detailed textures and colors, which also demonstrates that BTMnet can still achieve impressive results when dealing with bidirectional noise.

To compare the spectral restoration capabilities of different denoising methods, we provide the spectral curves of the denoising results obtained by each method on the WDC dataset (Case 6) in Fig. 7 with the wavelength bands plotted on the horizontal axis and the normalized pixel values on the vertical axis. A smoother spectral curve that closely resembles the ground truth indicates better spectral restoration performance. Fig. 7(a) shows the spectral curve of the noisy image, while Fig. 7(m) presents the spectral curve of the clean image, noise disrupts the original smoothness of the spectral curve. Fig. 7(b)–(l) depicts the spectral curves for various methods. Most of the comparative methods can restore the trend of the spectral curve but exhibit some degree of smoothness issues. The spectral curve of our proposed BTMnet is the smoothest and closest to the ground truth, which demonstrates BTMnet's exceptional spectral restoration capability.

D. Real Experiments

1) ZH-1 Dataset: The experimental results on the ZH-1 dataset for the bands (18, 24, 32) are shown in Fig. 8. The ZH-1 dataset contains Gaussian noise and wide stripe noise,

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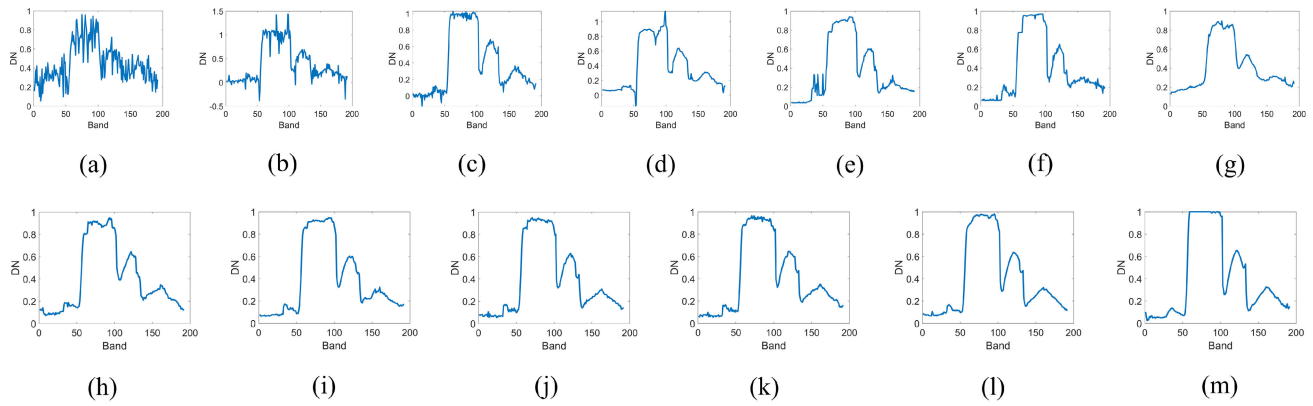


Fig. 7. Spectral curves of pixel (125, 80) for the WDC dataset (Case 5). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet. (m) Ground truth.

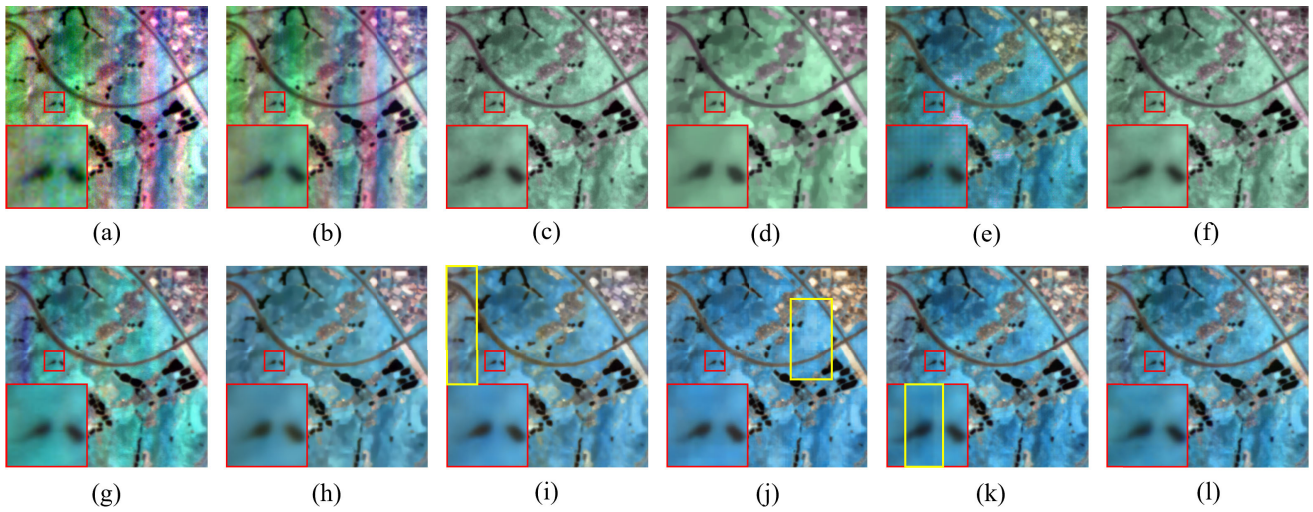


Fig. 8. Denoising results for the ZH-1 dataset in the real data experiment with bands (18, 24, 32). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet.

which cannot be eliminated effectively by most existing methods. LRMR, QRNN3D, LRTV, LRTDTV, GRNET, and MACNET all exhibit significant spectral distortion due to the diffusion of wide stripes into surrounding areas during processing. The restored image of SST exhibits significant mosaic blurring. As can be seen in Fig. 8(i), some stripes have not been removed on the far left of the display area in DSTrans. Fig. 8(k) shows a blue stripe on the SERT that has not been removed. While T3SC performs well in removing wide stripes and restoring spectral information, it falls short in restoring details. In contrast, the proposed BTMnet method not only effectively removes wide stripes but also preserves the spectral correlation of the image, restoring the original details of the image.

Additionally, we provide the horizontal digital number (DN) curve of the image, as shown in Fig. 9. In Fig. 9(a), the curves of the three wavelength bands displayed are all subject to significant fluctuations due to the interference of stripe noise. The results show that LRMR can only handle a small portion of mild stripes. The curves of LRTV, LRTDTV, and MACNET are overly smooth, resulting in the loss of most of the image details. The curves of GRNET and QRNN3D are not sufficiently flat, indicating that stripe residuals still exist. Although the curves of T3SC and SST are relatively smooth, the trend of the red curve does not match the original image, suggesting that they fail to preserve the original image information well. The blue and green curves of SERT are not as smooth as those of BTMnet, leading to some slight stripe residuals in the visual effect. Overall, the DN curve of the restoration results obtained by the proposed BTMnet is smoother while preserving the original image details, which also demonstrates that our method has a better effect on processing wide stripe noise.

2) GF-5 Dataset: The experimental results on the GF-5 dataset for the bands (160, 135, 95) are shown in Fig. 10. The original image is heavily disturbed by a large amount of Gaussian noise and wide stripe noise, especially the wide stripe noise. LRMR performs well in removing Gaussian noise but fails to eliminate wide stripe noise effectively. The denoising results of LRTDTV and GRNET are overly smooth, resulting in the loss of many details. MACNET, T3SC, DSTrans, and SST all exhibit significant spectral distortion due to the presence of wide stripe noise. Both QRNN3D

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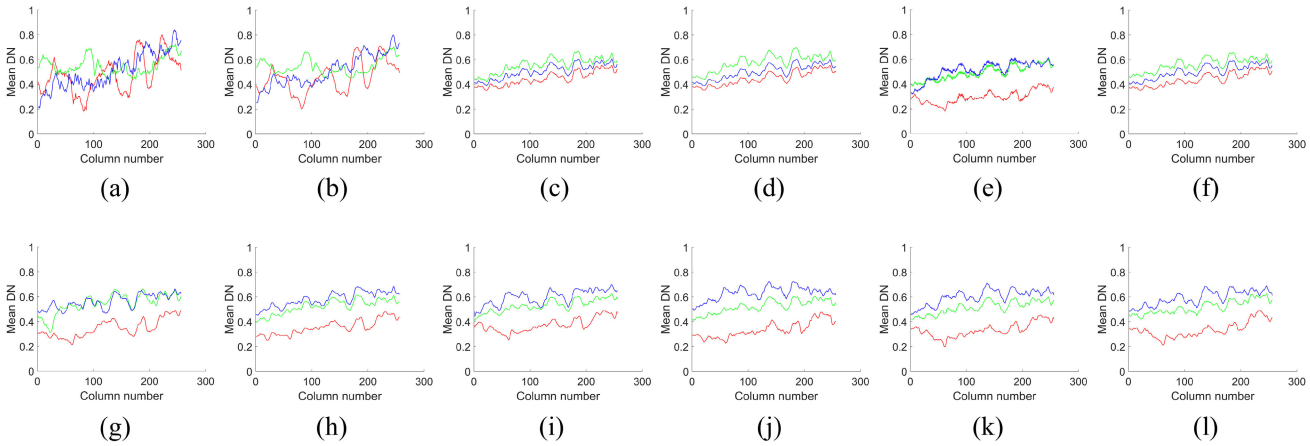


Fig. 9. Denoising results for the ZH-1 dataset in the real data experiment, displayed in horizontal mean DN curves of the band (red: 18, green: 24, blue: 32). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet.

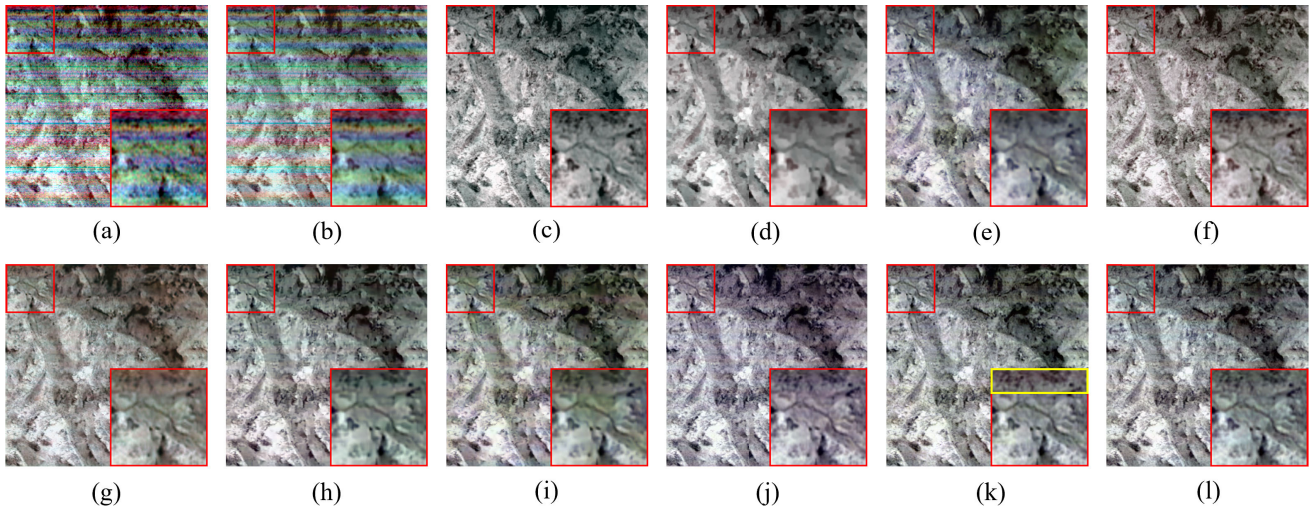


Fig. 10. Denoising results for the GF-5 dataset in the real data experiment with bands (160, 135, 95). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet.

and SERT have some residual stripes. In Fig. 10(k), a red wide stripe can be observed in the result of SERT that has not been completely removed. LRTV has a better effect in removing wide stripes but still lags behind BTMnet in detail restoration. Overall, the proposed BTMnet stands out among all comparison methods, not only effectively removing wide stripe noise but also demonstrating excellent detail restoration capabilities.

We also provide the vertical DN curve of the image, as shown in Fig. 11. Among the DN curves of all the comparison methods, the curves of LRTDTV and BTMnet are relatively smooth. However, based on the comprehensive visual inspection, the detailed restoration of LRTDTV is not satisfactory. In contrast, the proposed BTMnet exhibits the smoothest DN curve.

3) EO-1 Dataset: The experimental results on the EO-1 dataset for the bands (166, 129, 96) are shown in Fig. 12. The real imagery from EO-1 is severely affected by a combination of stripe noise, Gaussian noise, and impulse noise, making it difficult to remove the noise completely due to their intertwining nature. Among all methods, LRMR performs poorly in dealing with complex stripe noise. LRTV, LRTDTV, and MACNET suffer from significant spectral distortion due to the excessive presence of stripe noise. GRNET exhibits local spectral distortion, which may be attributed to its ineffective extraction of global features. QRNN3D, T3SC, DSTrans, SST, and SERT all have unremoved stripes in the locally enlarged images we provided, resulting in residual stripes in the visual results. In contrast, our method demonstrates superior denoising capability even in scenarios with complex noise.

4) Urban Dataset: The experimental results on the urban dataset for the bands (106, 87, 76) are shown in Fig. 13. Horizontal and vertical discontinuous stripe noises can be observed in the real imagery of urban. Among all the comparison methods, LRMR, LRTV, MACNET, and T3SC all fail to remove the horizontal stripe noise completely. The processing result of SST exhibits significant spectral distortion, possibly due to the diffusion of red stripes to the surrounding areas, resulting in a reddish appearance in the enlarged region. Although LRTDTV, GRNET, and SERT perform well in

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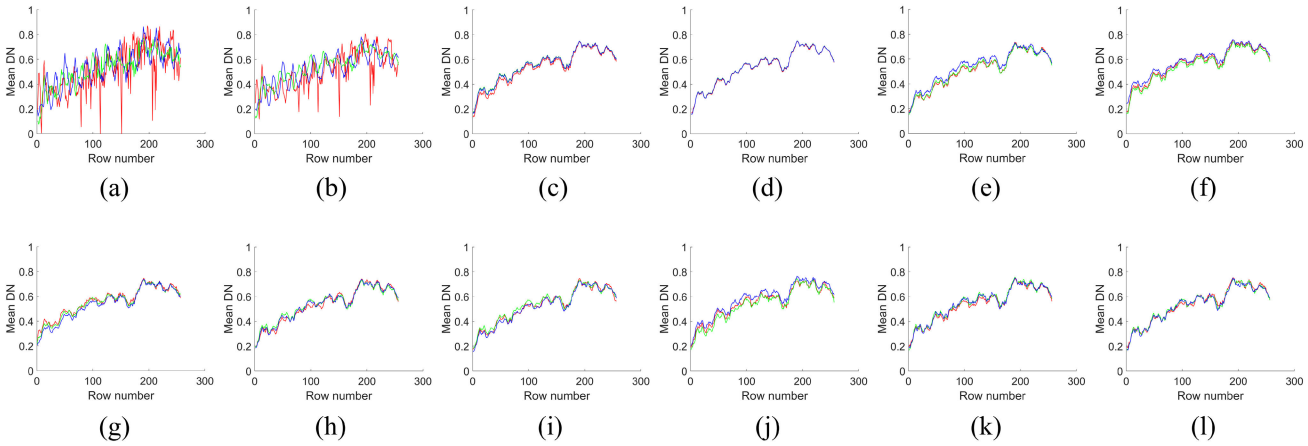


Fig. 11. Denoising results for the GF-5 dataset in the real data experiment, displayed in vertical mean DN curves of the band (red: 160, green: 135, blue: 95). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet.

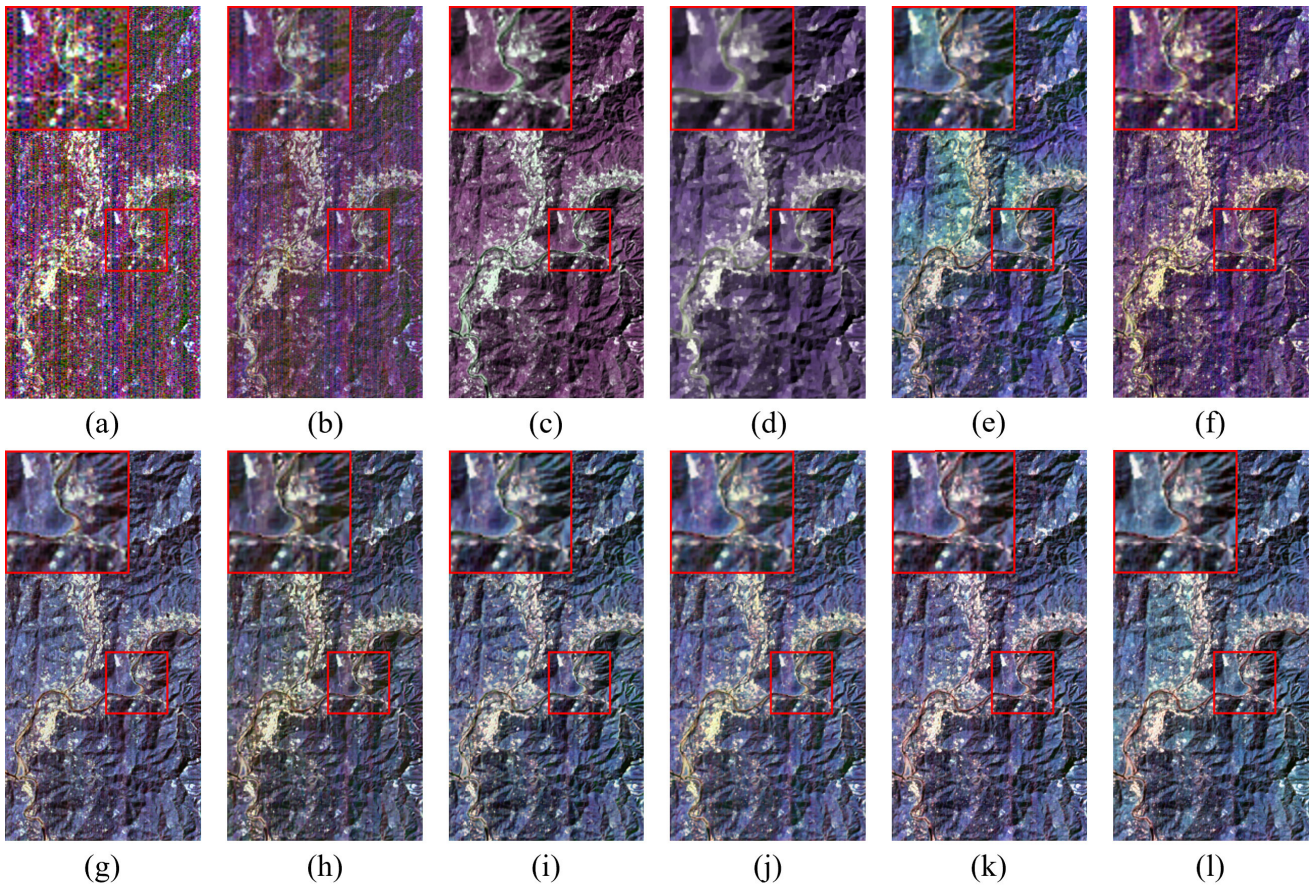


Fig. 12. Denoising results for the EO-1 dataset in the real data experiment with bands (166, 129, 96). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet.

stripe removal, they fail to restore the original details of the image. Overall, the proposed BTMnet demonstrates good performance in removing bidirectional discontinuous stripe noise, restoring details, and preserving spectral fidelity.

Additionally, to further leverage the ground truth of the HYDICE dataset, we applied SVM classification to the denoised results obtained by different methods. Initially, we randomly selected 10% of the labels from each of the six categories (asphalt, grass, tree, roof, metal, and dirt) to train. Then, the remaining 90% was used as the test set. Table II presents the quantitative metrics for the classification of the denoised results by different methods. Since not all bands of the Urban dataset are affected by noise, the overall accuracy (OA) and Kappa coefficient of the noisy image still reach 78.52% and 0.7209. Among all the compared methods, our BTMnet achieved an accuracy of 81.94% and

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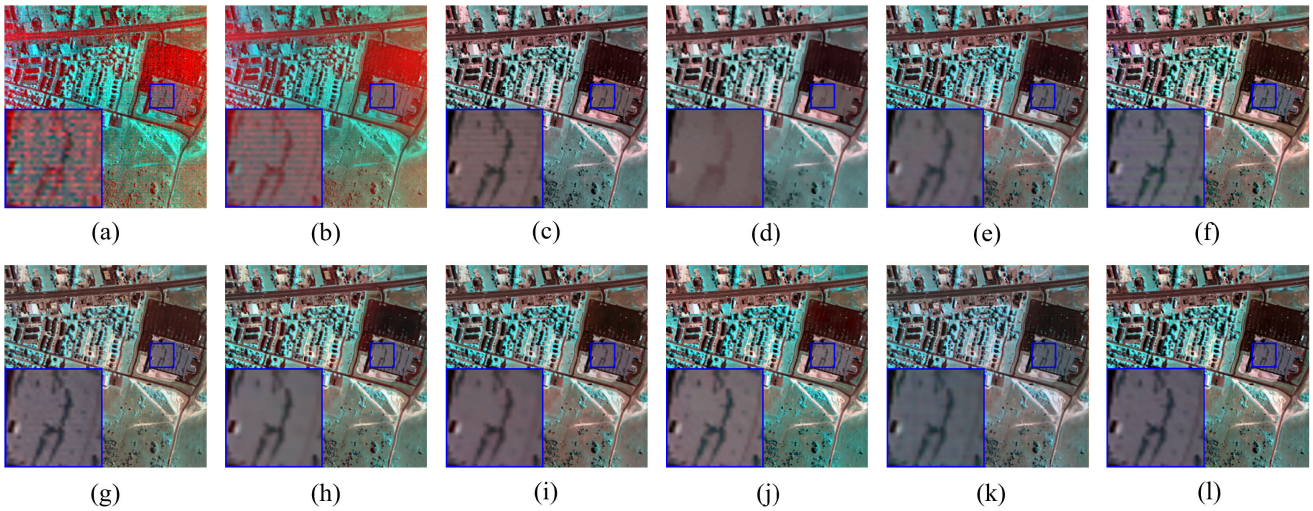


Fig. 13. Denoising results for the Urban dataset in the real data experiment with bands (106, 87, 76). (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet.

TABLE II
CLASSIFICATION ACCURACY RESULTS FOR THE URBAN DATASET IN THE REAL DATA EXPERIMENT

	Noise	LRMR	LRTV	LRTDTV	GRNET	MACNET	QRNN3D	T3SC	DSTrans	SST	SERT	BTMnet
OA↑	78.52%	76.78%	68.62%	65.09%	77.66%	80.67%	78.38%	77.95%	77.23%	79.49%	78.55%	81.94%
Kappa↑	0.7209	0.6980	0.5940	0.5531	0.7093	0.7485	0.7192	0.7139	0.7043	0.7338	0.7209	0.7645

a Kappa coefficient of 0.7645. Compared to the second-best method MACNET and the noisy image, BTMnet significantly outperforms them in both metrics. To visually compare the classification results, Fig. 14 provides the classification visualization outcomes of the denoised Urban dataset using different methods. It can be observed that our method is closer to the ground truth, especially in the classification of trees. In the top-right corner of the classification map, there are a large number of trees in the ground truth, and only the denoised result of our proposed BTMnet correctly identifies this area. Both quantitative metrics and visualization results have demonstrated that our proposed BTMnet can effectively remove noise from HSIs, restore more texture details, and consequently yield better classification results.

E. Ablation Study

We also conducted various ablation experiments in Case 5 on WDC to verify the effects of different network modules and to identify the optimal hyperparameters.

1) Impact of Different Network Modules: In Table III, we present the denoising performance of BTMnet after removing different network modules. Removing any one of the three modules will weaken the model's denoising capability. Among the three modules, the removal of SDA results in the most significant drop in network accuracy, with mPSNR decreasing by 2.02 dB, highlighting the crucial role of extracting local correlations in noise removal. Following closely is the CSM, the absence of this module results in a significant decrease in the mPSNR by 1.62 dB. This result indicates that combining local features with global features can further enhance the network's denoising performance. After removing LDA, the network's mPSNR drops by 0.66 dB, demonstrating the importance of utilizing LDA for interaction between local windows.

TABLE III
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT NETWORK CONSTRUCTION IN CASE 5 ON WDC TEST DATA

	mPSNR↑	mSSIM↑	SAM↓
w/o CSM	32.59	0.9673	5.28
w/o LDA	33.55	0.9716	4.95
w/o SDA	32.19	0.9641	5.29
BTMnet	34.21	0.9747	4.68

2) Impact of Bidirectional Windows: In Table IV, we present a comparison of the denoising performance of our proposed method when using single-direction windows versus bidirectional windows. The bidirectional windows evidently yield better denoising results compared to single-direction windows. Our design of bidirectional windows can simultaneously accommodate the extraction of features in both horizontal and vertical directions, thereby exhibiting superior denoising performance.

3) Impact of Different Window Shapes: In Table V, we present the performance of the network under different combinations of window sizes. To extract multiscale features, we adopt windows of various sizes in different TMGs, and as the network depth increases, the window sizes gradually enlarge. The window shapes listed in the table represent the two side lengths of rectangular windows. To determine the most suitable set of window sizes, we gradually

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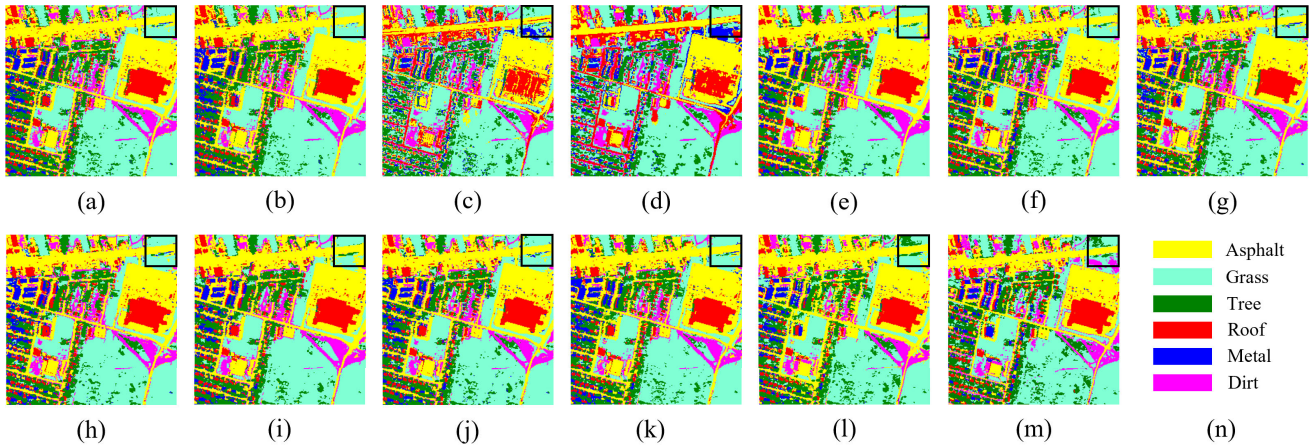


Fig. 14. Classification results for the Urban dataset in the real data experiment. (a) Noisy image. (b) LRMR. (c) LRTV. (d) LRTDTV. (e) GRNET. (f) MACNET. (g) QRNN3D. (h) T3SC. (i) DSTrans. (j) SST. (k) SERT. (l) BTMnet. (m) Ground truth. (n) Six classes.

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF BIDIRECTIONAL WINDOWS IN CASE 5 ON WDC TEST DATA

	mPSNR↑	mSSIM↑	SAM↓
w/o horizontal window	33.38	0.9691	5.07
w/o vertical window	33.06	0.9696	5.16
BTMnet	34.21	0.9747	4.68

TABLE V
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT WINDOW SHAPES IN CASE 5 ON WDC TEST DATA

Window Size			mPSNR↑	mSSIM↑	SAM↓
Group 1	Group 2	Group 3			
[16,4]	[16,8]	[16,16]	32.98	0.9670	5.15
[32,4]	[32,8]	[32,16]	32.30	0.9665	5.51
[64,4]	[64,8]	[64,16]	34.21	0.9747	4.68
[64,8]	[64,16]	[64,32]	33.75	0.9728	4.93

TABLE VI
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT DEPTHS OF NETWORK IN CASE 5 ON WDC TEST DATA

Layers	mPSNR↑	mSSIM↑	SAM↑	Params
2	32.32	0.9653	5.53	1.37M
3	34.21	0.9747	4.68	1.87M
4	33.79	0.9735	4.91	2.37M

TABLE VII
QUANTITATIVE EVALUATION RESULTS OF BIDIRECTIONAL SCANNING IN CASE 5 ON WDC TEST DATA

	mPSNR↑	mSSIM↑	SAM↓
w/o horizontal scanning	32.99	0.9680	5.30
w/o vertical window	32.84	0.9706	5.25
BTMnet	34.21	0.9747	4.68

TABLE VIII
QUANTITATIVE EVALUATION RESULTS OF THE CSM AND VANILLA MAMBA IN CASE 5 ON WDC TEST DATA

	mPSNR↑	mSSIM↑	SAM↓	Params
Vanilla Mamba	34.06	0.9770	4.70	2.96M
CSM	34.21	0.9747	4.68	1.87M

increase the window sizes in the three TMGs and ultimately find that the network performs best when the window sizes in the three TMGs are [64, 4], [64, 8], and [64, 16]. This set of windows is particularly effective at extracting features of wide stripe noise, and further increasing the window size would lead to more information redundancy.

4) Impact of Depths of the Network: In Table VI, we present the denoising performance of BTMnet with varying network depths. In the ablation experiment, we incrementally increased the number of network layers from 2 to 4. It can be observed that our proposed network achieves optimal performance when the number of layers is 3. Although this leads to an increase in the number of parameters compared to a network with two layers, the mPSNR improves by 1.89. When the number of layers is increased to 4, the network becomes more complex while the information becomes more redundant. This redundancy hinders the network training process. As a result, all three metrics experience a decline.

5) Impact of Bidirectional Scanning: In Table VII, we present a comparison of the denoising performance of our proposed method when using single-direction scanning in the CSM versus bidirectional scanning. Bidirectional scanning evidently obtains better denoising results compared to single-direction scanning. The design of bidirectional scanning can simultaneously achieve the extraction of features in both horizontal and vertical directions, thereby exhibiting superior denoising performance.

6) Impact of Channel-Split Scanning in the CSM: In Table VIII, we present a comparison of the denoising effects of the BTMnet before and after replacing the CSM with the vanilla Mamba in Vim [39]. Although the mSSIM increased by 0.0023 after the replacement, the mPSNR decreased by 0.15 and the SAM decreased by 0.02. Overall, CSM can enhance the performance of our model. Furthermore, our CSM employs a channel-split strategy, which avoids the redundancy

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of scanning in four directions across all channels. Consequently, the number of parameters is significantly reduced.

V. Conclusion

This article introduces a BTMnet for HSI denoising, named BTMnet. The BTMnet combines the powerful feature extraction capability of the Transformer with the lightweight global information extraction capability of Mamba. We propose two key modules: BLSDA and CSM. BLSDA extracts local features within windows and interactive features between windows both horizontally and vertically. The CSM extracts global information from the image through scanning in four directions and further integrates window features. Finally, we present experimental results to demonstrate the effectiveness of the BTMnet on both simulated and real datasets. We particularly highlight the excellent ability of the BTMnet to remove stripe noise, especially wide stripe noise. Additionally, we conduct in-depth discussions on the effectiveness of each module within the BTMnet and analyze the sensitivity of the network's window shapes. Although the stripes in our experimental data are primarily horizontal or vertical, HSIs in the real world contain stripes of various angles. In future research, we will strive to address more complex and realistic noise, including diagonal stripes.

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